PVT Behaviour, Enthalpy, Entropy and Fugacity of Pure Gases from Cubic Equations of State

Computer project report, Chemical Engineering Thermodynamics

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Program: viraj-chauhan.github.io/cubic-eos-calculator Source code: github.com/viraj-chauhan/cubic-eos-calculator

1. Objective

The project was given in three parts: (i) write a program that predicts the PVT behaviour of real gases with a cubic equation of state, (ii) extend it to calculate enthalpy and entropy, and (iii) add the fugacity of a pure component. We wrote a single program that does all three for the van der Waals (vdW), Redlich–Kwong (RK), Soave–Redlich–Kwong (SRK) and Peng–Robinson (PR) equations. The user picks one of 20 gases and enters T and P; the program reports Z, V, density, φ, f, H^{R}, S^{R}, H, S and the saturation properties for all four equations side by side. It runs in a web browser and needs no installation.

2. Theory

All four equations are special cases of the generic cubic of Smith et al. [1], which differ only in the constants σ, ε, Ω, Ψ and the function $\alpha(T_r)$ listed in Table 1:

$$P = \frac{RT}{V-b} - \frac{a(T)}{(V+\varepsilon b)(V+\sigma b)}, \quad b = \Omega\frac{RT_c}{P_c}, \quad a(T) = \Psi\frac{\alpha(T_r;\omega)\,R^2T_c^{\,2}}{P_c} \tag{3.41, 3.44, 3.45}$$

EOS	$\alpha(T_r)$	$\mathrm{d}\ln\alpha / \mathrm{d}\ln T_r$	σ	ε	Ω	Ψ	Z_c
vdW	1	0	0	0	1/8	27/64	3/8
RK	$T_r^{-1/2}$	−1/2	1	0	0.08664	0.42748	1/3
SRK	$[1+m(1-T_r^{1/2})]^2$	$-m\,T_r^{1/2}/[1+m(1-T_r^{1/2})]$	1	0	0.08664	0.42748	1/3
PR			$1+\sqrt{2}$	$1-\sqrt{2}$	0.07780	0.45724	0.30740

Table 1. Constants of the four equations (Table 3.1 of [1]); $m_{SRK} = 0.480 + 1.574\omega - 0.176\omega^2$, $m_{PR} = 0.37464 + 1.54226\omega - 0.26992\omega^2$.

With $\beta = bP/RT$ and $q = a/bRT$ (Eqs. 3.46, 3.47), substituting $V = ZRT/P$ into Eq. (3.41) gives a cubic in Z:

$$Z^3 + [(\sigma+\varepsilon-1)\beta - 1]Z^2 + [\sigma\varepsilon\beta^2 - (\sigma+\varepsilon)\beta(\beta+1) + q\beta]Z - [\sigma\varepsilon\beta^2(\beta+1) + q\beta^2] = 0 \tag{1}$$

This is the same equation as the iterative forms (3.48) and (3.49) in the book, written as a polynomial. The residual properties and the fugacity coefficient follow from Chapter 13 of [1]:

$$I = \frac{1}{\sigma-\varepsilon}\ln\frac{Z+\sigma\beta}{Z+\varepsilon\beta} \ \ (\text{vdW: } I=\beta/Z), \quad \ln\varphi = \frac{G^{R}}{RT} = Z - 1 - \ln(Z-\beta) - qI, \quad f = \varphi P \tag{13.72, 13.74, 10.34}$$

$$\frac{H^{R}}{RT} = Z - 1 + \left[\frac{\mathrm{d}\ln\alpha}{\mathrm{d}\ln T_r} - 1\right]qI, \qquad \frac{S^{R}}{R} = \ln(Z-\beta) + \frac{\mathrm{d}\ln\alpha}{\mathrm{d}\ln T_r}qI \tag{13.75, 13.76}$$

Total properties are found by adding the ideal-gas change from a reference state (ideal gas at $T_0 = 298.15$ K, $P_0 = 1$ bar, where $H = S = 0$) to the residual part, using $C_P^{ig}/R = A + BT + CT^2 + DT^{-2}$ (Eq. 4.4, constants from Table C.1):

$$H = R\int_{T_0}^{T}\frac{C_P^{ig}}{R}\mathrm{d}T + H^{R}, \qquad S = R\int_{T_0}^{T}\frac{C_P^{ig}}{R}\frac{\mathrm{d}T}{T} - R\ln\frac{P}{P_0} + S^{R} \tag{6.41}$$

The integrals are evaluated analytically. Below T_c, vapour and liquid are in equilibrium when $f^{v} = f^{l}$, i.e. $\varphi^{v} = \varphi^{l}$ (Eqs. 10.39, 10.41); this condition fixes P^{sat}, and the enthalpy of vaporization is then $\Delta H^{lv} = H^{R,v} - H^{R,l}$.

3. Program

The program is one file, `index.html`, with the calculation written in JavaScript. Keeping it in a single file means it can be opened offline or hosted directly on GitHub Pages. The calculation functions are kept separate from the code that draws the page:

Function	Task
`getParams`	$\alpha(T_r)$, a, b, q and d ln α/d ln T_r for the chosen EOS (Table 1)
`solveCubic`	all real roots of Eq. (1)
`solveEOS`	physical root(s), choice of stable root, V, ln φ, H^R/RT, S^R/R
`satPressure`	P^{sat}, saturated Z and V of both phases, ΔH^{lv}
`idealGas`, `calcAll`	ideal-gas H and S from C_P, and the totals H, S, f, density

Table 2. Main functions in the program.

Roots of the cubic

Newton–Raphson is started at $Z_0 = 1 + \max(|c_2|, |c_1|, |c_0|)$, which is an upper bound on every root of Eq. (1). From there the iteration decreases steadily onto the largest root. That root is divided out to leave a quadratic, whose two roots (if real) are refined by a few more Newton steps. Roots with $Z \le \beta$ are rejected because they give $V \le b$. If three roots remain, the smallest is the liquid and the largest the vapour; the middle one is unphysical. The program then compares ln φ for the two outer roots and reports the one with the lower value, since it has the lower Gibbs energy. The user can also force the liquid or vapour root to look at metastable states.

Saturation pressure

We solve $g = \ln \varphi^l - \ln \varphi^v = 0$ for ln P. At constant T, d ln φ/d ln $P = Z - 1$, so $g' = Z^l - Z^v$ and the Newton step is $\Delta \ln P = g/(Z^v - Z^l)$. The starting value comes from $\ln(P^{sat}/P_c) = 5.373(1 + \omega)(1 - 1/T_r)$ (Wilson's estimate). The program keeps a bracket on ln P and bisects whenever a Newton step leaves it or lands where the cubic has only one root. It stops when $|g| < 10^{-12}$; this usually takes 3 to 6 iterations.

Data

C_P constants are from Table C.1 of [1]. For the nine gases that also appear in the sample program shared in class (CH_4, C_2H_6, C_3H_8, n-C_4H_{10}, N_2, O_2, CO_2, NH_3, H_2O), T_c, P_c and ω are the same as in that program, so the two can be compared directly. The other eleven are taken from Appendix B of [1]. $R = 83.14$ cm^3 bar mol^{-1} K^{-1} is used in the EOS and 8.314 J mol^{-1} K^{-1} for H and S.

4. Results

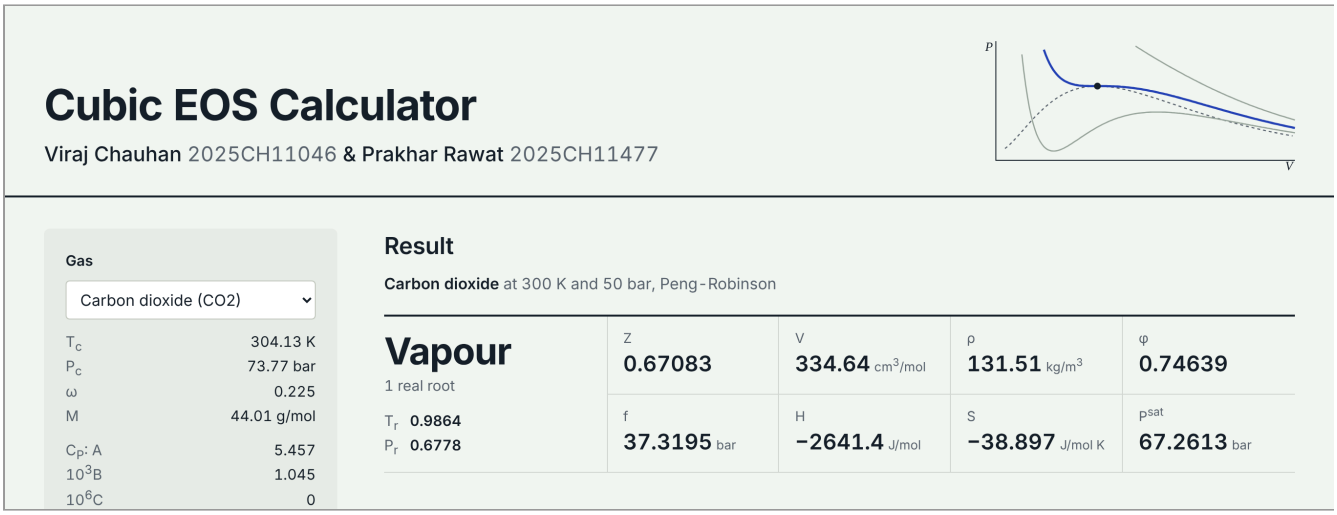


Figure 1. The program for CO_2 at 300 K and 50 bar. Inputs are on the left and the summary for the selected EOS (here PR) on the right; the comparison tables for all four equations follow below it on the page.

Table 3 shows the output for CO_2 at 300 K and 50 bar. This temperature is just below T_c ($T_r = 0.986$) and the pressure is below P^{sat}, so all four equations give a single vapour root. The gas is far from ideal: Z is 0.67–0.73 and the fugacity is 37–40 bar rather than 50 bar.

EOS	Z	V cm^3/mol	ρ kg/m^3	φ	f bar	H^R J/mol	S^R J/mol K	H J/mol	S J/mol K
Ideal gas	1	498.84	88.22	1	50.00	0	0	68.8	−32.294
vdW	0.73120	364.75	120.66	0.79241	39.62	−1672.8	−3.642	−1604.0	−35.936
RK	0.69714	347.76	126.55	0.76681	38.34	−2299.4	−5.457	−2230.5	−37.751
SRK	0.69433	346.36	127.06	0.76542	38.27	−2646.1	−6.598	−2577.3	−38.892
PR	0.67083	334.64	131.51	0.74639	37.32	−2710.2	−6.602	−2641.4	−38.897

Table 3. CO_2 at 300 K and 50 bar (reference state: ideal gas at 298.15 K, 1 bar).

To check the three-root logic, the same isotherm was run at 66 and 68 bar with PR, which predicts P^{sat} = 67.26 bar. At 66 bar the roots are Z = 0.20829, 0.25195 and 0.46920; the vapour root has the lower fugacity (43.95 bar against 44.15 bar for the liquid root), so the vapour is reported as stable. At 68 bar the roots are 0.20002, 0.32346 and 0.40382 and the order reverses (liquid 44.42 bar, vapour 44.53 bar), so the program switches to the liquid, V = 73.37 cm^3/mol. The switch happens exactly at P^{sat}, as it should.

Table 4 lists the saturation results for three fluids and compares them with experimental data. Water at 100 °C is compared with the steam tables in Appendix E of [1].

Fluid, T	EOS	P^{sat} bar	Z^l	Z^v	V^l cm^3/mol	V^v cm^3/mol	ΔH^{lv} kJ/mol
CO_2, 300 K (T_r = 0.986)	vdW	69.83	0.29074	0.46539	103.85	166.23	1.757
	RK	68.32	0.23336	0.44777	85.20	163.48	3.024
	SRK	67.39	0.22416	0.46001	82.96	170.25	3.913
	PR	67.26	0.20174	0.43507	74.81	161.33	3.954
	Expt. [2]	*67.13*					
Propane, 300 K (T_r = 0.811)	vdW	17.36	0.09911	0.73842	142.39	1060.9	7.303
	RK	11.52	0.04673	0.80205	101.16	1736.2	13.176
	SRK	10.09	0.03983	0.82314	98.44	2034.3	14.844
	PR	9.98	0.03472	0.81503	86.76	2036.6	14.750
	Expt. [2]	*9.97*					
Water, 373.15 K (T_r = 0.577)	vdW	15.18	0.01906	0.92173	38.95	1883.4	16.716
	RK	2.941	0.00250	0.97905	26.39	10329	33.756
	SRK	0.928	0.00076	0.99225	25.36	33164	42.961
	PR	0.965	0.00070	0.99173	22.50	31889	42.053
	Steam tables [1]	*1.013*	*0.00061*	*0.98441*	*18.81*	*30139*	*40.66*

Table 4. Saturation properties from the four equations compared with experimental values.

5. Validation

We checked the program in three ways. First, against the worked examples in [1]. Example 3.9 asks for the saturated vapour and liquid volumes of *n*-butane at 350 K and 9.4573 bar; Example 13.5 asks for Z, H^R and S^R of *n*-butane at 500 K and 50 bar with RK. The book's constants for *n*-butane were used for this check.

Quantity	Book	Program	Quantity	Book	Program
Ex. 3.9 V^v, vdW	2667	2666.9	V^l, vdW	191.0	190.98
V^v, RK	2555	2555.3	V^l, RK	133.3	133.27
V^v, SRK	2520	2520.3	V^l, SRK	127.8	127.81
V^v, PR	2486	2486.4	V^l, PR	112.6	112.60
Ex. 13.5 Z	0.6850	0.6852	H^R/RT	−1.0838	−1.0832
			S^R/R	−0.78735	−0.78687

Table 5. Comparison with worked examples in [1] (volumes in cm^3/mol).

The volumes agree to the last figure given in the book. The small differences in Example 13.5 come from the book rounding β and q to four or five figures before using them; our program carries full precision.

Second, we ran the sample program shared in class and our program on the same 20 states (nine gases, temperatures from 100 to 700 K, pressures from 5 to 300 bar, including liquid, vapour and supercritical states). For all four equations Z, V, φ, f, P^{sat}, H^R, S^R, H and S were identical to the precision printed. Third, as an internal consistency check, H^R was also obtained numerically from $H^R = -RT^2(\partial \ln \varphi/\partial T)_P$ by finite differences; it matched Eq. (13.75) to better than 0.01 J/mol for all four equations.

6. Discussion

- *Temperature dependence of a.* In vdW the attraction parameter does not change with T, so d ln α/d ln T_r = 0 and the energy terms are badly underestimated. For CO_2 at 50 bar, vdW gives H^R = −1673 J/mol against −2710 J/mol from PR, and for water at 100 °C it gives ΔH^{lv} = 16.7 kJ/mol against the steam-table value of 40.66 kJ/mol. RK improves this with $\alpha = T_r^{-1/2}$, and SRK and PR (42.1 and 43.0 kJ/mol) are within 6%.
- *Vapour pressure.* SRK and PR were fitted to vapour-pressure data through ω, and it shows: for propane they give 10.09 and 9.98 bar against 9.97 bar measured, while vdW is 74% too high. For water, a strongly polar fluid, PR is still within 5% (0.965 against 1.013 bar), but vdW and RK are far off.
- *Liquid volume.* Each cubic has a fixed Z_c (0.375, 0.333, 0.307) that is larger than the real one (0.229 for water, 0.274 for CO_2). All four therefore overpredict V^l; PR does best because its Z_c is lowest, but it is still 20% high for water (22.50 against 18.81 cm^3/mol). Vapour volumes are much better: PR is within 6% for water.
- *Near the critical point* (CO_2 at T_r = 0.986) the four equations disagree most, because ΔH^{lv} and the difference between V^l and V^v both go to zero at T_c and are sensitive to small errors in the EOS.

7. Conclusion

The program solves all four cubic equations of state and gives the volumetric properties, fugacity, enthalpy and entropy of a pure gas, including the saturation pressure and enthalpy of vaporization. Its results agree with the worked examples in the textbook and with the sample program, and the SRK and PR results agree well with experimental vapour pressures. Of the four equations, PR gives the best overall results for the fluids we tested, particularly for liquid volumes.

References

[1] J. M. Smith, H. C. Van Ness, M. M. Abbott and M. T. Swihart, *Introduction to Chemical Engineering Thermodynamics*, 8th ed., McGraw-Hill, 2018. Chapters 3, 4, 6, 10 and 13; Appendices B, C and E.

[2] E. W. Lemmon, M. O. McLinden and D. G. Friend, “Thermophysical Properties of Fluid Systems”, in *NIST Chemistry WebBook*, NIST Standard Reference Database 69, National Institute of Standards and Technology.